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HYBRID HUMAN-AI MATCHING SYSTEM WITH ACTIVE LEARNING

BACHELOR THESIS

5th Semester Report

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ABSTRACT

HYBRID HUMAN-AI MATCHING, ACTIVE LEARNING, SKILLED TRADES RECRUITMENT, LATVIA, WHATSAPP, NATURAL LANGUAGE PROCESSING

This 5th semester bachelor thesis report presents progress on the development of a Hybrid Human-AI Matching System with Active Learning, addressing the skilled trades recruitment challenge in Latvia and the Baltic region. The report describes the problem statement, thesis objectives, results of an analysis of relevant literature and existing systems, and a development schedule for the remaining thesis work.

The system combines AI-driven candidate screening with human oversight, using WhatsApp voice-based onboarding to lower the barrier for skilled trades workers with limited digital literacy. An active learning feedback loop enables the system to continuously improve its matching quality based on recruiter input.

The bachelor thesis 5th semester report consists of 15 pages; it contains 1 table and 23 references.

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INTRODUCTION

Latvia and the Baltic states face a pronounced and worsening shortage of qualified skilled trades professionals including electricians, plumbers, welders, carpenters, and HVAC technicians. This structural gap is driven by aging workforce demographics, outmigration of skilled labour following EU accession in 2004, and an educational system that has historically underinvested in vocational pathways (Hazans, 2011). The State Employment Agency of Latvia consistently ranks trades positions among the ten hardest to fill occupations in the country.

Traditional recruitment methods are poorly suited to reaching trades workers, who rely on informal referral networks rather than online job boards or resume portals (Sproge, 2020). Existing AI-powered recruitment platforms are designed for knowledge work and are largely inaccessible to trades workers. Research shows that fully automated AI recruitment systems can introduce systematic bias, further motivating a hybrid approach in which humans remain in the decision loop (Raghavan et al., 2020). This thesis addresses the gap by developing a Hybrid Human-AI Matching System that uses WhatsApp voice-based onboarding and an active learning feedback loop to deliver accessible, regulation-compliant recruitment support for small and medium-sized enterprises in Latvia.

The goal of the bachelor thesis is to develop and evaluate a hybrid human-AI platform for skilled trades matching in Latvia that uses WhatsApp voice-based onboarding and an active learning feedback mechanism to improve both accessibility and matching quality over time. The tasks of the bachelor thesis are:

1. To analyse literature and existing research on the challenges of skilled trades recruitment in the Latvian and Baltic labour market.
2. To research hybrid human-AI architectures and existing similar systems.
3. To design a hybrid human-AI architecture for intelligent candidate and job matching.
4. To implement a system prototype including voice-based profile extraction, a matching engine, and a feedback mechanism.

5. To integrate an active learning component to enable continuous improvement based on human feedback.
6. To evaluate the system effectiveness in terms of matching quality, user satisfaction, and learning performance compared to a static AI approach.
7. To discuss ethical implications, limitations, and potential impact in the Latvian context.

Chapter 1 presents the problem statement, justification of topicality, and development schedule. Chapter 2 presents the results of analysis of information sources across four thematic areas: the Baltic skilled trades labour market, hybrid human-AI architectures, active learning methodology, and voice-based AI recruitment tools. The report concludes with results and conclusions summarising the findings of the semester.

1. LABOUR MARKET CHALLENGES AND THESIS OBJECTIVES

The statement of the problem. Latvia and the Baltic states are experiencing a pronounced and worsening shortage of qualified skilled trades professionals. According to data from the State Employment Agency of Latvia, construction and trades positions consistently constitute a high proportion of the hardest to fill vacancies. This structural gap is exacerbated by aging workforce demographics, outmigration of skilled labour to Western European markets following EU accession, and an educational system that has historically underinvested in vocational pathways (Hazans, 2011).

Traditional recruitment methods including job boards, newspaper advertisements, and employment agency intermediaries are poorly suited to reaching skilled trades workers, who tend to rely on personal referral networks. Existing AI-powered recruitment platforms have been designed predominantly for knowledge work, with features such as resume parsing and online profile integration that are largely irrelevant or inaccessible to the trades sector. The result is a persistent mismatch between employer demand and labour supply, inflicting economic costs on SMEs in the construction, maintenance, and infrastructure sectors (Sproge, 2020).

The goal of the bachelor thesis. To develop and evaluate a hybrid human-AI platform for skilled trades matching in Latvia that uses WhatsApp voice-based onboarding and an active learning feedback mechanism to improve both accessibility and matching quality over time.

Justification of topicality. The topicality of this thesis is grounded in three converging trends: the structural labour market gap in Latvian skilled trades, the rapid advancement of large language models and speech recognition APIs that make voice-based AI assistants commercially viable, and growing academic and industry interest in hybrid human-AI workflows that leverage complementary strengths of machine efficiency and human judgment (Grosz et al., 2022).

The European Commission 2023 Skills Agenda identifies construction trades as a critical shortage occupation across multiple member states, including Latvia. The National Development Plan of Latvia 2027 identifies workforce development in construction and energy sectors as a strategic priority. Despite this, most technological investment in HR technology has flowed to sectors with higher educated, digitally literate workforces.

WhatsApp has near universal penetration among working-age adults in Latvia and supports voice message functionality that many trades workers already use daily for informal coordination. By meeting workers in the communication channels they already inhabit, the system avoids the onboarding friction that defeats most digital recruitment platforms (Montag et al., 2015). Active learning has been shown to dramatically reduce labelled data requirements for classification tasks (Settles, 2012). In a recruitment context, this means a system that learns to distinguish good matches from poor ones using a relatively small number of validated recruiter judgements rather than requiring large historical datasets.

The hybrid architecture, in which AI pre-screens and ranks candidates but humans make final decisions and provide feedback, addresses concerns about algorithmic bias and regulatory compliance. The EU AI Act, which entered into application in 2024 to 2025, classifies AI systems used in employment and worker management as high risk, imposing requirements for transparency, human oversight, and non-discrimination. A system designed from the outset for human-in-the-loop operation is better positioned to meet these requirements (Raghavan et al., 2020).

Table 1.1

Bachelor Thesis Development Schedule

Activity	Deadline
Collect and survey literature sources	Week 3
Analyse information sources and draft Chapter 2	Week 6
Submit 5th semester progress report	Week 9
Design system architecture	Week 10
Implement WhatsApp voice onboarding module	Week 11

Implement matching engine and active learning loop	Week 12
Pre-defence	Week 12
Conduct evaluation and write results chapter	Week 15
Write ethical implications and conclusions	Week 16
Bachelor thesis submission	Week 17 (Tuesday)
Bachelor thesis defence	Week 19

2. ANALYSIS OF INFORMATION SOURCES

The analysis of information sources for this thesis draws on peer-reviewed academic literature, industry reports, technical documentation, and policy documents. Sources were identified through Google Scholar, IEEE Xplore, and the ACM Digital Library. The following sections summarise the key findings organised by thematic area: the skilled trades recruitment landscape in the Baltic region; hybrid human-AI architectures for recruitment; active learning methodology; and voice-based and conversational AI in recruitment contexts.

2.1. Skilled trades recruitment in the Baltic labour market

The labour market dynamics of Latvia and the broader Baltic region are characterised by structural supply and demand imbalances that disproportionately affect skilled trades occupations. Hazans (2011) provides a comprehensive longitudinal analysis of Latvian labour migration, documenting the emigration of approximately 200,000 working-age adults in the decade following EU accession, with disproportionate outflows from construction and manufacturing sectors. This demographic drain has left a persistent gap in the domestic supply of qualified tradespeople.

The State Employment Agency of Latvia publishes quarterly occupational demand forecasts that consistently rank electricians, welders, construction labourers, and HVAC technicians among the top ten hardest to fill positions. Unlike graduate-level occupations where shortages can be partly addressed by attracting international talent, trades positions require language competency and often involve physical presence at specific job sites, limiting the effectiveness of international recruitment.

Eglite and Miežaine (2018) analyse the structural causes of trades shortages from an education systems perspective, arguing that the collapse of vocational training infrastructure in the post-Soviet transition period has not been adequately reversed despite two decades of EU-funded reform efforts. The authors document a particular weakness in apprenticeship pathways, which in Germany and Austria

account for over 50 percent of new skilled trades entrants but in Latvia represent fewer than 5 percent of secondary school graduates.

Sproge (2020) examines the recruitment practices of Latvian construction SMEs through a qualitative study of 47 firms employing between 5 and 50 workers. The study finds that 78 percent of successful hires are made through personal referral networks, with formal channels accounting for fewer than 15 percent of hires. This finding is consistent with broader literature on semi-skilled labour markets, where informal information networks serve a gatekeeping function that is difficult for algorithmic systems to replicate (Granovetter, 1973).

The Baltic Labour Studies Institute regional report extends this analysis to Estonia and Lithuania, finding similar patterns of informal network dependence in trades hiring, though with country-specific variations (BLSI, 2022). Estonian firms report somewhat higher adoption of digital recruitment tools, attributed to the country's broader e-governance infrastructure, while Lithuanian firms show recruitment patterns most similar to Latvia's SME-dominated construction sector.

Sondore and Zarina (2023) examine the language dynamics of the Latvian trades labour market, noting that a substantial portion of available skilled workers are Russian-speaking and may face additional barriers when engaging with Latvian-language recruitment platforms. This multilingual dimension reinforces the argument for a voice-based interface that can handle multiple languages naturally, and informs the NLP pipeline design decisions discussed in Section 2.4.

Critically, none of the reviewed studies identify a digital platform that has achieved meaningful penetration in Baltic skilled trades hiring. The absence of a validated solution suggests that the problem may be one of product design and channel strategy rather than fundamental technical infeasibility. The literature consistently points toward mobile-first, voice-accessible tools as the most promising avenue for reaching this demographic.

2.2. Hybrid human-AI architectures for candidate matching

The concept of hybrid human-AI systems, also termed human-in-the-loop AI or augmented intelligence, has received substantial attention in both academic AI research and industry practice. Amershi et al. (2019) provide an influential taxonomy of human-AI interaction patterns, distinguishing systems in which humans override AI decisions after the fact, systems that solicit human input during processing, and systems that use human feedback to improve future AI performance. The proposed thesis system integrates all three patterns.

In the specific domain of recruitment AI, Raghavan et al. (2020) conducted an audit study of commercial AI hiring tools, finding systematic disparate impact across gender and race dimensions in screening algorithms. Their analysis highlights why purely automated recruitment systems face both technical and regulatory challenges: the training data for such systems tends to reflect and amplify historical hiring biases. The authors recommend human oversight as a necessary safeguard, particularly for initial screening decisions that function as gatekeepers for subsequent stages.

Grosz et al. (2022) examine human-AI collaboration in high-stakes decision environments and propose a framework for calibrating the division of labour between algorithmic and human components. Their key insight is that AI should handle volume and consistency while humans handle edge cases and contextual judgement. This maps naturally onto a recruitment scenario where AI pre-screens a large candidate pool and human recruiters make final evaluations of shortlisted candidates.

Leavy (2018) surveys bias mitigation strategies in natural language processing systems relevant to recruitment, including debiasing word embeddings, adversarial training approaches, and counterfactual data augmentation. The review concludes that no single technique reliably eliminates bias across diverse demographic groups and recruitment contexts, reinforcing the case for human oversight in the decision loop.

Koechling and Wehner (2020) conducted a systematic review of AI in recruiting and selection, analysing 75 peer-reviewed papers published between 2014 and 2019. They identify accuracy, efficiency, and fairness as the three central dimensions on which AI recruiting systems are evaluated, and note that fairness considerations are frequently under-addressed in system design. Their findings further

motivate the hybrid design of this thesis, where human judgment provides a fairness safeguard that purely algorithmic systems cannot guarantee.

Srivastava et al. (2022) demonstrate that large language model based systems can perform well on domain-specific matching tasks without task-specific training when prompted appropriately. This finding has direct relevance to the matching engine design: rather than training a bespoke classifier from scratch on scarce Latvian trades data, the system may leverage a pre-trained language model with carefully crafted prompts as an initial matching mechanism, reserving active learning for fine-tuning on recruiter feedback.

2.3. Active learning feedback mechanisms

Active learning is a machine learning paradigm in which a learning algorithm iteratively selects the most informative data points for labelling by a human annotator, with the goal of maximising model performance under labelling budget constraints. The foundational framework is described by Settles (2012), whose comprehensive survey remains the definitive reference for the field. The survey distinguishes among query by committee strategies, uncertainty sampling, expected model change, and query by disagreement, each appropriate for different learning scenarios.

Uncertainty sampling, in which the model queries instances for which its current predictions are least confident, is the most computationally tractable approach and the most widely validated in practice (Settles, 2012). For a binary matching task of good candidate versus poor candidate, uncertainty sampling maps naturally to requesting human review of candidate and job pairs for which the model assigns a probability close to 0.5.

Zhang et al. (2019) apply active learning to job and candidate matching specifically, using a neural representation of job requirements and candidate profiles. Their experiments demonstrate that active learning significantly outperforms passive learning in the early stages of deployment when labelled data is scarce. Critically, they find that the choice of feature representation, particularly the quality of semantic embeddings, matters more than the specific active learning query strategy, suggesting

that investment in the NLP pipeline is more impactful than fine-tuning the query mechanism.

Monarch (2021) provides a practitioner-oriented treatment of active learning for NLP tasks, with particular attention to the challenges of deploying active learning in production systems rather than experimental benchmarks. The author identifies annotation fatigue as a key challenge. The system design must therefore limit the number of feedback requests per recruiter session and ensure that individual queries are quick and natural to answer.

Hu et al. (2020) demonstrate that neural language representations can generalise across languages, enabling a system trained on English language job descriptions to be applied to Latvian language candidate profiles with acceptable performance degradation through cross-lingual transfer. Their XLM-R model achieves strong performance on multilingual benchmarks without task-specific fine-tuning, making it a viable starting point for the matching engine before recruiter feedback has been accumulated.

Bangert et al. (2023) introduce a stream-based active learning approach for rapidly evolving job market data, demonstrating that pool-based active learning strategies can be adapted to handle the temporal dynamics of job postings and candidate availability. Their approach is particularly relevant for the Latvian seasonal construction market, where demand patterns shift substantially between summer and winter months and a static model would quickly become outdated.

2.4. Voice-based and conversational AI in recruitment

The use of voice interfaces and conversational AI in recruitment has grown substantially since the commercialisation of accurate automatic speech recognition systems around 2017 to 2018. Manyika et al. (2017) estimate that voice-based interaction will account for a growing share of human and computer interaction, particularly in use cases where text input is inconvenient, precisely the situation facing a skilled trades worker on a construction site or in a workshop.

WhatsApp voice message feature is of particular relevance to this thesis. Unlike phone calls, which require synchronous availability, voice messages allow asynchronous communication that candidates can send during breaks or after work. Unlike text messages, they require no typing and make language-specific keyboard installations unnecessary. Montag et al. (2015) find that WhatsApp voice messaging is disproportionately popular among users with lower formal education levels and among older age groups, precisely the demographic of experienced trades workers.

The technical pipeline for converting WhatsApp voice messages to structured candidate profiles involves several components: audio format conversion, automatic speech recognition, language identification, named entity recognition for extracting skills and certifications, and dialogue management for follow-up questions. Radford et al. (2023) introduce the Whisper model, which demonstrates near-human performance on multilingual speech recognition across 99 languages including Latvian and Russian. This substantially reduces the technical risk of the voice onboarding component.

For the dialogue management component, Bocklisch et al. (2017) present the Rasa framework for open-source conversational AI, which provides tools for both natural language understanding and dialogue policy learning. The framework support for multiple languages through custom tokenisation and its open-source licence make it appropriate for the proposed system. Alternatively, recent large language model based dialogue systems can handle multi-turn conversations with minimal fine-tuning, potentially simplifying the architecture.

Maurer et al. (2023) evaluate the performance of language model powered conversational agents for structured information extraction from unstructured speech, finding that chain-of-thought prompting substantially improves extraction accuracy for domain-specific entities such as trade certifications and work experience periods. Their approach is directly applicable to the voice onboarding component of this thesis, where the goal is to extract a structured candidate profile from a free-form voice narrative.

Sondore and Zarina (2023) evaluate speech recognition quality for Latvian language speech on current open-source models, reporting that Whisper large-v3 achieves a word error rate of approximately 8 percent on spontaneous Latvian speech. For Russian language input, error rates are even lower at approximately 5 percent, confirming that the speech recognition component can handle both major language communities in the Latvian trades workforce without requiring custom model training.

Ivanov and Webster (2019) study candidate experience with AI-driven interaction in hiring processes, finding that candidates are sensitive to perceived fairness when interacting with AI systems. This reinforces the importance of transparent communication about the system role and limitations in the onboarding dialogue design, and supports the hybrid approach where human recruiters remain visibly in the loop throughout the process.

RESULTS AND CONCLUSIONS

The analysis of information sources conducted during the 5th semester has produced the following results and conclusions, which form the foundation for the design and implementation work to be carried out in the final thesis.

First, the literature confirms a clearly identified and unmet need in the Latvian and Baltic skilled trades labour market. No existing digital platform has achieved meaningful penetration in this segment, and the reviewed studies consistently identify informal voice-based communication and personal networks as the dominant channels through which trades workers find employment (Sproge, 2020; BLSI, 2022). This validates the core design hypothesis that a WhatsApp voice-based interface is the appropriate channel for candidate onboarding.

Second, the review of hybrid human-AI architectures confirms that a human-in-the-loop design is both technically sound and appropriate under the EU AI Act regulatory framework. The literature documents systematic fairness failures in fully automated recruitment systems (Raghavan et al., 2020; Koechling & Wehner, 2020), and the proposed hybrid approach addresses these failures directly. The framework proposed by Grosz et al. (2022) will serve as the primary architectural reference for the system design.

Third, the active learning literature establishes that uncertainty sampling with cross-lingual neural embeddings, specifically XLM-R as demonstrated by Hu et al. (2020), is the most appropriate starting point for the matching engine given the multilingual nature of the Latvian labour market and the expected scarcity of labelled recruiter feedback data in the early deployment phase. The temporal adaptation approach of Bangert et al. (2023) will be incorporated to handle seasonal demand variation.

Fourth, the evaluation results reported by Sondore and Zarina (2023) confirm that Whisper large-v3 meets the accuracy requirements for the voice onboarding pipeline in both Latvian and Russian, eliminating the need for a custom speech recognition model and reducing the technical risk of the thesis. The chain-of-thought

prompting approach from Maurer et al. (2023) will be integrated into the voice processing pipeline.

The planned next steps for the final thesis semester are to proceed with system architecture design, implementation of the WhatsApp voice onboarding module, development of the matching engine with active learning integration, and empirical evaluation of the complete prototype with real recruiter feedback from Latvian SMEs. The development schedule presented in Table 1.1 will guide this work.

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